

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC2-133***

***Warren County 500 kV
20 MW Capacity / 20 MW Energy***

May 2018

Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, Section 205, as well as the System Impact Study Agreement between Virginia Electric & Power Company, the Interconnection Customer (IC) and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the IC. As a requirement for interconnection, the IC may be responsible for the cost of constructing Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an IC may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The IC is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

The IC has proposed an 20 MW uprate to the Warren County Power Station natural gas generating facility located at 477 Kelley Drive in Front Royal, Virginia. The installed facilities will have a total capability of 1484 MW with 1370 MW of this output being recognized by PJM as capacity. See the table below for a summary of all the queue requests and uprates associated with the Warren County Power Station. The expected Commercial Operation Date for this uprate project is **June 1, 2018**.

Queue Position	Description	MFO (MW)	Capacity (MW)
V2-030	New 2 X 1 combined cycle unit	950	875
V4-018	Convert to 3 X 1 combined cycle unit	+475	+415
W3-047	PJM super peak operating configuration additional Capacity and energy.	+39	+60
AC2-133	Improve efficiency to increase output	+20	+20
Total	Total Plant Output and Capacity	1484	1370

Point of Interconnection

AC2-133 “Warren County 500 kV” uprate project will be an increase in Maximum Facility Output (MFO) and Capacity to the existing Warren County Power Station gas generation plant which interconnects with the ITO transmission system at the Front Royal 500 kV Switchyard. The single line is shown below in **Attachment 1**.

Cost Summary

The AC2-133 interconnection request will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 0
Total Costs	\$ 0

Attachment Facilities

Generation Substation: None.

Transmission: None.

Direct Connection Cost Estimate

Substation: None.

Transmission: None.

Non-Direct Connection Cost Estimate

Substation: None.

Transmission: None.

Interconnection Customer Requirements

ITO's Facility Interconnection Requirements as posted on PJM's website:

<http://www.pjm.com/~media/planning/plan-standards/private-dominion/facility-connection-requirements1.ashx>

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Interconnected Transmission Owner Requirements

Metering and SCADA/Communication equipment must meet the requirements outlined in section 3.1.6 Metering and Telecommunications of ITO's Facility Connection Requirement NERC Standard FAC-001 which is publically available at www.dom.com.

Network Impacts

The Queue Project AC2-133 was evaluated as a 20.0 MW (Capacity 20.0 MW) uprate to the Warren County units in the ITO area. Project AC2-133 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC2-133 was studied with a commercial probability of 100%. Potential network impacts were as follows:

PJM assessed the impact of the proposed Queue Project as an injection into the ITO, for compliance with NERC Reliability Criteria. The system was assessed using the summer 2020 RTEP case. When performing analysis, ITO Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under single contingency (normal and stressed system conditions). A full listing of the ITO's Planning Criteria and interconnection requirements can be found in the ITO's Facility Connection Requirements which are publicly available at: <http://www.dom.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically NERC Category C Contingency Conditions (Bus Fault, Tower Line, N-1-1, and Stuck Breaker scenarios) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For ITO Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating. The results of these studies are discussed in more detail below.

Summer Peak Analysis - 2020

Base Case Used

Summer Peak Analysis – 2020 Case

Contingency Descriptions

The following contingencies resulted in overloads:

None.

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None

Contributions to previously identified circuit breakers found to be over-duty:

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None.

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

Stability: **No mitigations** were found to be required.

Reactive Power Design Criteria:

The existing MW portion of the Front Royal facility shall retain its existing ability to maintain a power factor of at least 0.95 leading to 0.90 lagging measured at the generator's terminals.

The increases related to AC2-133 shall be designed with the ability to maintain a power factor of at least 1.0 (unity) to 0.90 lagging measured at the generator's terminals.

Reactive Power Assessment:

The queue project was found to be **compliant** for the MFO values requested for Summer and Winter based on the suggested operating point and the project design capability.

Specific requirements are summarized in **Attachment 2**, Tables 1 to 4. The design capability curves for the units are provided in **Appendix A**. Available reactive power was obtained based on design capability curve information.

For the CT's, PJM used 39.3 degrees cooling water temperature for summer and 20 degrees cooling water temperature for winter for reactive capability calculations.

Affected System Analysis & Mitigation

Duke, Progress & TVA Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The IC can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

System Reinforcements

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None.

Light Load Analysis - 2020

Light Load Studies to be conducted as required by PJM Manual 14B.

Not required.

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

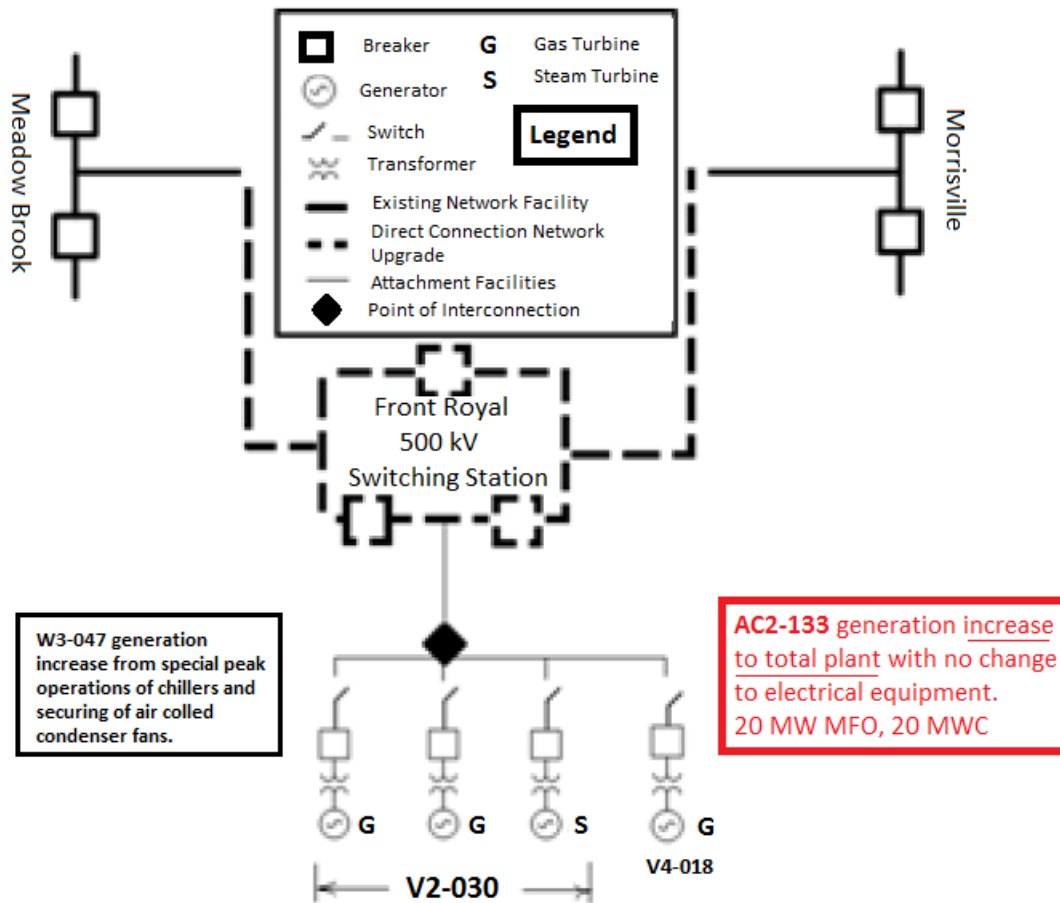
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None.

Attachment 1.

System Configuration



Attachment 2.

Reactive Power Assessment

MFO Reactive Capability (Winter):

Table 1. AC2-133: Reactive Power Assessment for each CT G1, G2, G3 (MFO values)

AC2-133 CT		350	MVA		
		Required PF range			
	MW	Lagging	Leading	MAX MVAR	MIN MVAR
Existing MW (Gross MFO)	308	0.9	0.95	149	101
AC2-133 increase (Gross)	2	0.9	1	0.96	0.00
Total Requirements (MFO)	310			149.96	101
Avail reactive power at 310 MW*				165	124
Total deficiency in MVAR				-	-

*The available reactive power capability is determined using the design capability curve of the machine at the machine gross active power output.

Table 2. AC2-133: Reactive Power Assessment for ST (MFO values)

AC2-133 ST		682	MVA		
		Required PF range			
	MW	Lagging	Leading	MAX MVAR	MIN MVAR
Existing MW (Gross MFO)	572	0.9	0.95	277	188
AC2-133 increase (Gross)	18	0.9	1	8.71	0.00
Total Requirements (MFO)	590			285.71	188
Avail reactive power at 590 MW*				340	247
Total deficiency in MVAR				-	-

*The available reactive power capability is determined using the design capability curve of the machine at the machine gross active power output.

MFO Reactive Capability (Summer):

Table 3. AC2-133: Reactive Power Assessment for each CT G1, G2, G3 (MFO values)

AC2-133 CT		318	MVA			
		Required PF range				
	MW	Lagging	Leading	MAX MVAR	MIN MVAR	
Existing MW (Gross MFO)	273	0.9	0.95	132.2	89.73	
AC2-133 increase (Gross)	2	0.9	1	0.96	0.00	
Total Requirements (MFO)	275			133.16	89.73	
Avail reactive power at 275 MW*				160	115	
Total deficiency in MVAR				-	-	

*The available reactive power capability is determined using the design capability curve of the machine at the machine gross active power output.

Table 4. AC2-133: Reactive Power Assessment for ST (MFO values)

AC2-133 ST	682	MVA			
		Required PF range			
	MW	Lagging	Leading	MAX MVAR	MIN MVAR
Existing MW (Gross MFO)	563	0.9	0.95	272.6	185
AC2-133 increase (Gross)	18	0.9	1	8.71	0.00
Total Requirements (MFO)	581			281.39	185
Avail reactive power at 581 MW*					
				357	250
Total deficiency in MVAR					
				-	-

*The available reactive power capability is determined using the design capability curve of the machine at the machine gross active power output.

Conclusions

No reactive deficiencies were identified for the Front Royal units with the addition of AC2-133 request.

APPENDIX A:

CT Reactive Curve

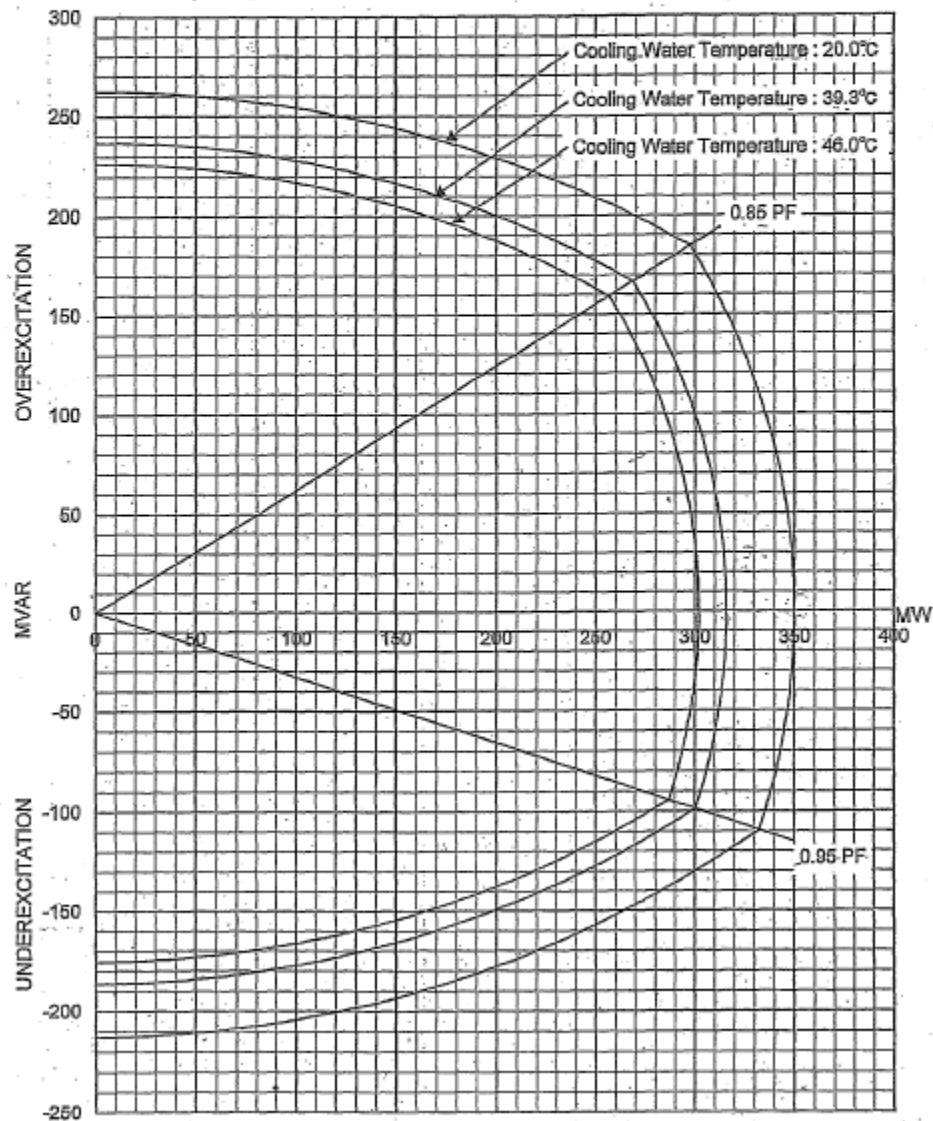
KC919702A

Dominion / Warren County C/C GTG



HYDROGEN INDIRECTLY COOLED GENERATOR CAPABILITY CURVE

350000 kVA, 297500 kW, 21 kV, 60 Hz, 3600 min⁻¹,
0.85 PF, 330 VEXC, 0.3 MPa-g GAS PRESSURE



ST Reactive Curve

KC919717B

Dominion / Warren County STG



WATER COOLED GENERATOR CAPABILITY CURVE

682000 kVA, 579700 kW, 19 kV, 60 Hz, 3600 min⁻¹,
0.85 PF, 500 VEXC, 0.5 MPa-g GAS PRESSURE

Cooling Water Temperature : Max. 39.3°C

